

Bowie State University
Department of Computer Science
College of Arts and Sciences
Fall 2026
COSC 473/573 (3 cr) ARTIFICIAL INTELLIGENCE (I)

INSTRUCTOR:

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Important Reminders from the Bowie State University Administration

REQUIRED TEXTBOOK:

Artificial Intelligence: A Modern Approach (Fourth Edition) by [Stuart Russell](#) and [Peter Norvig](#)

COURSE DESCRIPTION:

COSC 473 (Prerequisite - COSC 214): This course is an introduction to the basic concepts and techniques of artificial intelligence. Topics include: knowledge representation, search strategies, fuzzy and probabilistic reasoning, and theorem proving. Applications of AI languages and the design and construction of Expert Systems are discussed.

COSC 573 (Prerequisite – COSC 528): This course introduces students to the ideas, challenges, techniques, and problems in artificial intelligence. The course will cover both fundamental concepts such as search and knowledge representation, as well as applied work in areas such as planning and vision. Topics covered include basic search, heuristic search, game search, constraint satisfaction, knowledge representation, logic and inference, probabilistic modeling, and machine learning. Students are also exposed to the history and ethical questions related to artificial intelligence.

STUDENT LEARNING OBJECTIVES:

- **Understand** numerous aspects of computational models of intelligence
- **Analyze** and **discuss** AI concepts and techniques
- **Identify** AI algorithms suitable for different problems
- **Design** AI solutions and **implement** AI systems
- **Communicate** AI topics effectively through oral presentations and written reports
- **Become** an effective team member for successful group work

ATTEMPTED PROGRAM OUTCOMES:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
3. Communicate effectively in a variety of professional contexts.
6. Apply computer science theory and software development fundamentals to produce computing-based solutions. [CS]

COURSE OBJECTIVES:

- Ability to apply, compare and evaluate various search strategies
- Ability to represent knowledge in propositional logic and first order logic
- Ability to apply propositional and first order logic inference techniques
- Ability to apply AI techniques to find solutions to given problems
- Ability to analyze, implement and evaluate various solutions to AI problems
- Ability to work in teams to solve computational problems

STUDENT/COURSE OUTCOMES:

- **Understand** numerous aspects of computational models of intelligence
- **Identify** AI algorithms suitable for different problems
- **Implement** AI concepts and techniques
- **Design** and **Apply** AI-based solutions to solve practical problems
- **Communicate** AI topics effectively through oral presentations and written reports

INSTRUCTION MODES:

The course will be taught as a series of lectures, class discussions, class exercises, homework problems and a final programming project. The students are expected to read the material before class to take active participation in class discussions. Students are expected to write programs, create reports and make presentations as requested by the instructor.

COURSE POLICIES

- Follow the postings on Blackboard regularly.
- Be on time and take an active role in class discussions.
- Sign the attendance sheet to record your presence in the classroom.
- Solve and submit all assignments on time.
- Respect the rights of others seeking to learn.
- Silence all electronic devices that distract classroom learning.
- Maintain a professional outlook in speech and attire.
- Credit sources when graded work involves any type of collaboration

POLICY ON ATTENDANCE: Regular attendance in the class is mandatory. University rules and regulations state that a student may fail a course if the student misses a certain number of classes [six (6) for a course that meets three (3) times a week; four (4) for a class that meets twice a week; and two (2) for a class that meets once a week]. Students will be responsible for any loss of information, assignments, and projects due to absence from class.

POLICY REGARDING ACADEMIC INTEGRITY: Students are expected to conform to a strict standard of academic honesty. Students are required to do all work individually unless specified otherwise by the instructor. Collaboration resulting in plagiarism is considered cheating and will be subject to discipline. Cheating on examinations, plagiarism, unauthorized collaboration with others on assignments, submitting duplicate assignments for credit in more than one course without authorization, and improper acknowledgment of sources of material are intolerable offenses that carry serious penalties. Penalties may consist of a failing grade on the affected assignment, a failing grade in the course, or more severe discipline imposed by the University.

USE OF GENERATIVE AI IN THIS COURSE: In this course, you are permitted and encouraged to use artificial intelligence (AI) powered programs to help you with all work done outside the classroom. However, you should be aware that AI text generation tools may present incorrect information, biased responses, and incomplete analyses, which may result in answers that do not meet the standards of this course. It is your responsibility to verify the answers through your independent study before submitting for grading. Submitting a wrong answer will not get any credit. To adhere to our university values, you must cite any AI-generated material (e.g., text, images, and

other content) included or referenced in your work and provide the prompts used to generate the content. Using an AI tool to generate content without proper attribution will be treated as plagiarism and reported.

STUDENT GRADE APPEAL PROCEDURES: Students who have documentation that an error was made in the determination of a course grade or desire to appeal a grade due to alleged arbitrary and capricious grading **must first discuss the concern with the faculty member involved**. If there are unresolved issues, the student may initiate the grade appeal process. The student must direct the appeal in writing, with documentation, through the appropriate channels: Instructor, Department Chair, School Dean, and Provost. **If a course assignment grade is in dispute, the student must initiate the appeal no later than ten (10) working days after receiving the grade**. If a final grade is in dispute, the student must submit an appeal no later than thirty (30) days from the end of the semester during which the grade was received. Procedures are detailed below.

1. The student must initially consult with the instructor for a satisfactory resolution of the appeal. If the instructor is no longer with the University or if the student is unable to resolve the issue with the instructor, the student must present the complaint in writing to the Department Chair.
2. The Department Chair will convene a Departmental Appeals Committee for resolution. If the student is not satisfied with the decision of the Departmental Appeals Committee, the Chair renders a decision.
3. If the complaint cannot be resolved at the Departmental level, the student must present the complaint in writing to the School Dean within fifteen (15) days after receiving notification of the Department's decision.
4. The School Dean will render a final resolution.

I. ASSESSMENTS

Group assignments will be mainly in class with a few take-home components. The research project will involve doing research, coding, report writing and presentation. For each test, you can bring one cheat sheet of handwritten notes (not printed or photocopied) with your name on it. The size of the sheet cannot be over a letter size paper.

Rubric for programming assignments:

	Advanced 100	Proficient 75	Developing 50	Beginning 25
Correctness 40%	All outputs provided are correct.	Most outputs are correct.	Some outputs are correct	A very few outputs are correct
Readability 10%	Meaningful names for variables and functions. Properly commented. Modular design. Small units, no lengthy functions.	Mostly meaningful names, commented code separated into smaller units/functions.	Occasionally meaningful names, commented code separated into smaller units/functions	Rarely meaningful names, commented code separated into smaller units/functions
Completeness 40%	Program works as intended for all possible inputs.	Program works as intended for most inputs.	Program works as intended for some inputs	Program works as intended for a few inputs
Efficiency 10%	Program is very efficient in terms of number of operations and memory usage.	Program is mostly efficient in terms of number of operations and memory usage.	Program is moderately efficient in terms of number of operations and memory usage	Program is not adequately efficient in terms of number of operations and memory usage

Rubric for non-programming assignments:

Technical Dimensions/Weight	Exceeds Expectations (8-10)	Meets Expectations (4-7)	Does not Meet Expectations (1-3)
Overview 15%	Provides an engaging overview, thoroughly defines the scope, and gives a clear sense of purpose	Provides an adequate overview and a general explanation of the scope. Gives a reasonable sense of purpose	Fails to provide an overview and define the scope. Fails to give a reasonable sense of purpose
Concepts, Reasoning	Ideas and concepts are logical	Main concepts and ideas are	Confused about assignment and

25%	and work together as a unified whole. Main points are supported with specific evidence.	visible but not always logical and/or too broad. Support for main points is given but not always sufficient and/or focused.	relevant concepts. Lack of support for main points Illogical presentation of ideas
Soundness, Relevance 25%	All necessary analysis and/or sources of information are clearly stated. All results and references are valid, reliable, specific, and relevant to the assignment	Sufficient information and/or analysis is given but results are not always credible, reliable and/or relevant to the assignment	Analysis and/or sources of information are not clearly specified, not reliable and/or not relevant to the assignment
Completeness 20%	Followed all directions and responded to all parts of the assignment. Work is self-contained and complete with respect to the stated scope.	Most of the directions are followed and most parts of the assignment are addressed. Work covers most of the stated scope and is reasonably self-contained and complete.	Directions are not followed and/or work is incomplete and does not address one or more of the points necessary to treat the given topic
Originality 15%	Content clearly displays evidence of independent and critical thinking	Some ideas display capability of independent and critical thinking or perspective	Ideas are unrefined and/or fail to demonstrate independent and critical thinking

Rubric for group assignments:

Goal	4 Points	3 Points	2 Points	1 Point
Equal Work 20%	Did a full share of work or more	Did an equal share of work	Did almost as much work as others	Did little or no work
Cooperation 20%	Took an initiative in helping the group get organized	Worked agreeably with partners	Could be persuaded to cooperate	Did not cooperate
Participation 20%	Provided many ideas	Participated in discussions and made some suggestions	Listened to others but offered few suggestions	Seemed bored with the discussions and offered no suggestions
Support 20%	Assisted other partners	Offered encouragement to other partners	Seemed preoccupied with own project	Took little interest in others' projects
Communication 20%	Clearly communicated ideas	Usually, shared ideas	Rarely expressed ideas	Never expressed any ideas

II. GRADING FACTORS & GRADING SCALE

Homework is due on time. Assignments that are submitted late may be accepted at the discretion of the instructor. However, late assignments will be penalized and can receive only 80% of the original grade. All individual assignments carry equal weights. All group assignments carry equal weights. Both the tests carry equal weights. **All non-programming graded work** (except when explicitly stated in class) **need to be handwritten**!

The grades will be distributed as follows:

Class Participation and Group Assignments	15%
Individual Assignments	25%
Midterm	20%
Final Project, Research Paper and Presentation	20%
Final	20%

Letter grades will be awarded as follows:

- A: ≥ 90 and $\leq 100\%$
- B: ≥ 80 and $< 90\%$
- C: ≥ 70 and $< 80\%$
- F: $< 70\%$

TENTATIVE COURSE OUTLINE

*****ANY REVISIONS ARE LEFT TO THE DISCRETION OF THE INSTRUCTOR.*****

Session	Topic
1	Introduction, (Ch 1), Intelligent Agents (Ch 2)
2	Uninformed Search (Ch 3)

Session	Topic
3	Informed Search (Ch 3)
4	Local Search (Ch 4.1-4.4)
5	AI and Games –Adversarial Search (Ch 5.1 – 5.3)
6	Midterm Evaluation, Project Proposals Due
7	Constraint Satisfaction Problems (Ch 6)
8	Logic based agents (Ch 7 and 8)
9	Inference in First Order Logic (Ch 9)
10	Quantifying uncertainty (Ch 13)
11	Bayesian Networks (Ch 14.1-14.4)
12	Temporal Models (Ch 15.1-15.3)
13	Markov Decision Process (Ch 16.1-16.4, 17.1-17.3)
14	Machine Learning - Classification (Ch 18.1, 18.2, 18.6)
15	Learning Probabilistic models (Ch 20.1-20.2)
16	Final Evaluation

III.AMERICAN DISABILITY ACT (ADA) AND NONDISCRIMINATION POLICY

- Bowie State University is committed to creating inclusive and accessible learning environments consistent with federal and state law. If you have registered with Disability Support Services (DSS) and an accommodation notification was sent on your behalf, please schedule a meeting with me to discuss how your accommodation will be implemented in this course. Students who have a permanent disability or temporary health condition that requires accommodation should contact Disability Support Services immediately. You may email: dss@bowiestate.edu , call 301-860-4085 or stop by their office: Thurgood Marshall Library, lower-level, RM# 078. Our website provides guidance on how to register with Disability Support Services: <https://www.bowiestate.edu/academics/support-services/disability-support-services/request-services/> It is important that you reach out early to ensure adequate time to process your accommodation request and establish an approved plan, if necessary. Please note that approved accommodations are not retroactive.
- Bowie State University shall not discriminate against any individual on the basis of race, color, religion, age, ancestry or national origin, sex, sexual orientation, gender identity, disability, marital status, or veteran status.